

Mohamed Ayman Elmansy

Computer Engineering Graduate

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Professional Summary

Computer Engineering graduate from Kuwait University with a GPA of 3.96/4.00 and experience in software development, embedded systems, robotics, machine learning, computer networks, and database systems. Skilled in building practical engineering solutions that combine hardware, software, sensors, control systems, and web-based interfaces. Experienced with Python, Java, C++, C, SQL, JavaScript, Flask, Arduino, Raspberry Pi, ESP32, GitHub, Linux, and machine learning workflows. Award-winning competitive programmer and robotics participant with strong problem-solving, teamwork, documentation, and technical communication skills.

Education

Kuwait University

Bachelor of Engineering in Computer Engineering

GPA: 3.96/4.00

Shadadiya, Kuwait

Sep 2022 – Jun 2026

Relevant Coursework: Data Structures, Operating Systems, Network Programming, Computer Architecture, Signals and Systems, Probability and Statistics, Database Systems, Embedded Systems, Hardware Design, Machine Learning, Numerical Analysis, Wireless and Mobile Communication.

Technical Skills

Programming Languages: Python, Java, C++, C, SQL, JavaScript, HTML, CSS

Software Development: Object-Oriented Programming, Data Structures, Algorithms, Web Development, Socket Programming, Multithreading, REST APIs, Flask, Front-End Development, GitHub, Agile Fundamentals

Embedded Systems and Robotics: Arduino, Raspberry Pi, ESP32, Sensors, Motor Control, PWM, Servo Motors, Ultrasonic Sensors, IR Sensors, Bluetooth Control, Serial Communication, Robot Control Systems

Artificial Intelligence and Machine Learning: Machine Learning, Data Preprocessing, Classification, Regression, CNNs, Model Evaluation, Scikit-learn, Pandas, NumPy, Matplotlib, Google Colab

Databases and Tools: SQL, Oracle APEX, MS Access, GitHub, Linux, Netlify, Google Sheets Integration, LaTeX, Microsoft Office, Technical Documentation

Computer Engineering Areas: Computer Networks, Embedded Systems, Computer Architecture, Digital Systems, Cybersecurity Fundamentals, CUDA Programming, Signal Processing

Professional Experience

Engineering Graphics Instructor and Technical Tutor

Instructor and Technical Content Developer

Kuwait

2023 – Present

- Delivered academic support in engineering graphics and scientific subjects including mathematics, physics, and chemistry.
- Created structured explanations, visual learning materials, and technical content to simplify complex engineering and scientific concepts.
- Strengthened communication, teaching, documentation, and presentation skills through direct interaction with students.

Graphic Designer and Front-End Contributor

Freelance / Project-Based Work

Kuwait

2021 – Present

- Designed visual content, interface layouts, and presentation materials for organizations, companies, and online channels.
- Applied HTML, CSS, JavaScript, and responsive design principles to improve project interfaces and web-based prototypes.
- Used design experience to improve project reports, technical demonstrations, user interfaces, and presentation quality.

Selected Projects

Aquaculture Monitoring ROV for Kuwait Bay

Sep 2025 – Jun 2026

Capstone Project

- Designed an underwater ROV system for water-quality monitoring and sampling at 0–15 m depth, integrating sensors, camera monitoring, telemetry, and mission-based data logging.
- Defined requirements for 1 Hz sensor logging, depth control, sampling accuracy, leak detection, emergency stop, live telemetry, and automated post-mission reporting.

Autonomous Robot Control System with Web Dashboard

Apr 2025 – May 2025

Raspberry Pi, Arduino, Flask, Python

- Built a robot control system using Raspberry Pi and Arduino with manual driving, obstacle avoidance, live camera streaming, and real-time sensor feedback.
- Integrated 3 ultrasonic sensors, 3 IR tracking sensors, Bluetooth commands, serial communication, and a Flask dashboard for hardware-software control.

Network Programming Chatroom Application

May 2025 – Jun 2025

Java, Python, Flask, Sockets

- Developed a local-network chatroom application using sockets, multithreading, Java, Flask, JavaScript, and web technologies.
- Implemented client-server communication, concurrent user handling, message exchange, and desktop/web access over a local network.

Careema Ride-Hailing Prototype

Jun 2025 – Jul 2025

Software Engineering Project

- Designed and developed a female-to-female ride-hailing prototype focused on safety, usability, requirements analysis, and structured software design.
- Created system diagrams, user flows, UI testing materials, and a mobile-friendly web prototype using HTML, CSS, and JavaScript.

Modified Median Filter Using CUDA

Jul 2025 – Aug 2025

CUDA C, C, GPU Programming

- Implemented CPU and GPU versions of a modified median filter algorithm for image noise reduction and performance comparison.
- Applied CUDA kernels, parallel execution, memory management, and execution-time analysis to evaluate GPU acceleration.

Arabic Multimodal Lie Detection System

Mar 2026 – May 2026

AI, Machine Learning, Web Development

- Designed a data-collection and machine learning pipeline for Arabic truth and deception detection using audio, video, and metadata.
- Planned separate audio-based and visual-based model workflows with participant flow, consent handling, data storage, web deployment, and model demonstration.

Certifications

Introduction to IoT and Digital Transformation Credential: View Credential	Cisco Networking Academy
Introduction to Cybersecurity Credential: View Credential	Cisco Networking Academy
CCNA: Routing and Switching Credential: View Credential	Cisco Networking Academy

Honors and Awards

VEX Robotics Excellence Award VEX Robotics	Feb 2026
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- Received the VEX Robotics Excellence Award for strong overall achievement in robot design, programming, teamwork, technical documentation, and engineering problem-solving during the 2025–2026 competition season.

2nd Place Team – Kuwait Collegiate Programming Contest (KCPC) ACPC	Oct 2024
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- Achieved 2nd place in KCPC 2024 as part of a competitive programming team.
- Solved algorithmic problems under time constraints using data structures, dynamic programming, graph algorithms, mathematics, and efficient problem-solving techniques.

3rd Place Team – Kuwait Collegiate Programming Contest (KCPC) ACPC	Oct 2023
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- Achieved 3rd place in KCPC 2023 as part of a competitive programming team.
- Applied competitive programming techniques involving algorithms, mathematical reasoning, optimization, and fast implementation.

Languages

Arabic: Native | English: Advanced

Areas of Interest

Robotics, Embedded Systems, Software Engineering, Artificial Intelligence, Machine Learning, Computer Vision, Computer Networks, Autonomous Systems, Sensor-Based Platforms, Cybersecurity, Competitive Programming.